

Hydrogeochemical evaluation of groundwater in and around Lakkireddipalli and Ramapuram, Y.S.R District, Andhra Pradesh, India

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ABSTRACT

To evaluate hydrogeochemical evaluation of groundwater for drinking & irrigation practices in and around Lakkireddipalli and Ramapuram, Y.S.R District, Andhra Pradesh, India, thirty groundwater samples were collected during March 2018. Most of the groundwater samples are above the safe limits of W.H.O standards. Groundwater is exceeding the permissible limits of TDS, TH, EC, chloride and sulphates. Geogenic and anthropogenic sources are the main contributing factors for fluoride and nitrate respectively. Rock dominance is the predominant hydrogeochemical factor controlling the water chemistry. Piper's diagram reveals that water is of mixed Ca Mg Cl type. Irrigational water quality indices reveal that majority of groundwater samples falls under the unsuitable category and application of this untreated irrigational water is harmful for the crops and decreases the crop yield. Hence more saline tolerant crops could be suggested to combat soil salinization and application of lime/gypsum may be suggested to about the soil permeability.

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1. Introduction

Reliable and accessibility to safe drinking water is needed for sustainable development. Groundwater quality plays a major role in determining the usage of groundwater for a specific purpose and water for drinking purpose must be devoid of harmful elements, excessive minerals, living and nonliving organisms which may impact human health (Reddy Muralidhara et al., 2013). As public health directly linked with quality of drinking water, its systematic monitoring is highly needed. Water reliability for different purposes relies on the chemical and physical quality of water. Groundwater chemistry is governed primarily through both natural and anthropogenic variables. The lithological, geochemical composition was impacted by groundwater's hydrochemical features through subsurface movement (Elango et al., 2003). This underground water through pore spaces and weathered formations can change by the action of various different hydrochemical mechanisms (Rajmohan and Elango, 2004; Brindha and Elango, 2012). Groundwater chemistry can be altered through different anthropogenic sources which include point sources like waste disposal practices, hygienic conditions (Amoako et al., 2011; Brindha and Elango, 2015). Efficient evaluation of the physico-chemical parameters, their sources, the control of hydrogeochemical procedures are crucial for the sustainability of the ecosystem. Many scientists have researched physicochemical and

hydrochemical parameters to evaluate the groundwater characteristics (Subba Rao et al., 2002; Bhardwaj and Singh, 2011).

Groundwater is considered as a primary requirement owing to the unavailability of a proper source of water. Hence groundwater quality has changed as one of the most crucial environmental issue (Ravichandra and Chandana, 2006). Groundwater is the prime source of drinking water supply for both urban & rural population in India. Groundwater constitutes above 88% of safe drinking water in rural areas (Jain et al., 2010). Apart from drinking, groundwater is also used for irrigation & industrial processes. Advanced methodologies in pumping techniques have made a demand on groundwater in present days. As the surface water pollution is increasing rapidly the extraction of groundwater is also increasing considerably and further groundwater pollution & its misuse is becoming more significant. In India groundwater contributes nearly 80% of domestic practices and >45% of total agricultural water irrigating 39 million hectares (Pawar and Shaikh, 1995; Kinzelbach et al., 2003; Kumar et al., 2005; Sunitha et al., 2014b, 2014a). Hence systematic monitoring of groundwater is very essential. According to the previous literature groundwater pollution is associated with many sources like geogenic and anthropogenic, nitrogen pollutions agricultural practices dissolution of rocks and soil (Guo et al., 2007; Brindha and Elango, 2010; Elango et al., 2012; Brindha and Elango, 2014; Sunitha et al., 2012a; Sunitha et al., 2014b, 2014a; Adimalla and Venkatayogi, 2017; Sunitha and Muralidhara Reddy, 2018; Prasad et al., 2019). Decline of groundwater quality is of major concern not only in developed countries like India and China and also

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important worldwide (Narasimha and Jianhua, 2019). In India especially in Andhra Pradesh, groundwater quality studies with reference to drinking and irrigation purposes in different regions have been carried out, viz., Anantapur District (Sunitha et al., 2012a, 2012b, 2012c; Reddy Muralidhara et al., 2019); Kadapa District (Sunitha et al., 2013a, 2013b, 2016; Reddy Sudarshan et al., 2018); Visakhapatnam (Subba Rao and Krishna Rao, 1991; Subba Rao et al., 2012, 2016). Groundwater is the prime sources of water for drinking, agricultural needs semi-arid region of Y.S.R District. Geochemistry of groundwater & its sustainability for drinking, agricultural purpose hasn't yet studied in this region and hence this study was emphasized on an evaluation of suitability of water for drinking and irrigation practices.

2. Study area

Lakkireddy palli, Ramapuram Mandal of Y.S.R District, Andhra Pradesh lies between North latitude $14^{\circ}18'0''$ – $14^{\circ}7'30''$ and east longitude $78^{\circ}20'30''$ – $78^{\circ}49'30''$ with an area 510.272 km² and falls in Survey of India toposheet numbers 57 J16 and J12 (Fig. 1). The region is covered with different litho units types belonging to late Archean or Early Proterozoic era succeeded by rocks of Dharwarian age and both are traversed by dolerite dykes. Geology map is shown in Fig. 2. The older rocks are superposed by Cuddapah Supergroups of rocks and Kurnool group of middle-upper proterozoic age. Various lithological units observed are granites, granodiorites, granite-gneiss, quartzites, shales, limestones, dolerite dykes, and quartz reefs. Alluvium comprises of gravel, sand, silt, and clay occurs along with the Pennar River courses. The Thickness of the top zone up to 10 m in the study area. Groundwater occurs in fractured granitic rocks at greater depths. Fractured zone was found at different depths of ranging from 8 to 145.80 bgl. Potential fractures were noticed at 20 to 100 m bgl. The cumulative yield of fractured zones ranges between 0.1 and 4.9 lbs. However overall yield of bore wells sometimes is even higher depending on the fractured area in the

bore well. An annual average rainfall of 759 mm is noticed during Southwest monsoon (June to September) period (Central Ground Water Board, 2013). Red soil accounts for 53% of cultivated area.

3. Materials & methods

Thirty groundwater samples were collected from borewells in and around Ramapuram & Lakkireddy palli Mandals of Y.S.R District, Andhra Pradesh during March 2018. Sample locations are given in Fig. 1. In order to evaluate the suitability of water drinking, irrigation practices water quality parameters like pH, electrical conductivity, Total Dissolved Solids (TDS), Total Hardness, Ca^{2+} , Mg^{2+} , Na^{+} , K^{+} , CO_3^{2-} , HCO_3^{-} , Cl^{-} , SO_4^{2-} , NO_3^{-} , F^{-} and various irrigational quality indices like Percent Sodium (%Na), Sodium adsorption ratio (SAR), Residual Sodium Carbonate (RSC), Permeability Index (PI), Magnesium Ratio (MR), Chloroalkaline indices CA1, CA2, Kelly's ratio were used. The techniques and methods adopted for collection, parameter analytical techniques were followed according to Hem (1985), Raghunath (1987), Karanth (1989), and APHA (2005). The data is depicted in Table 1.

3.1. Analytical accuracy

To measure analytical accuracy between the concentration of total cations (Na^{+} , K^{+} , Ca^{2+} , Mg^{2+}) and the concentration of total anions (CO_3^{2-} , HCO_3^{-} , Cl^{-} , SO_4^{2-} , NO_3^{-} and F^{-}) expressed in milliequivalent per liter (meq/L) for each sample, ionic balance error (IBE) was checked to ensure the accuracy of analysis by the equation given below:

$$\text{IBE} = \frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \times 100$$

The computed ionic balance errors (IBE) were within the acceptable limit of $\pm 5\%$ (Domenico and Schwartz, 1990).

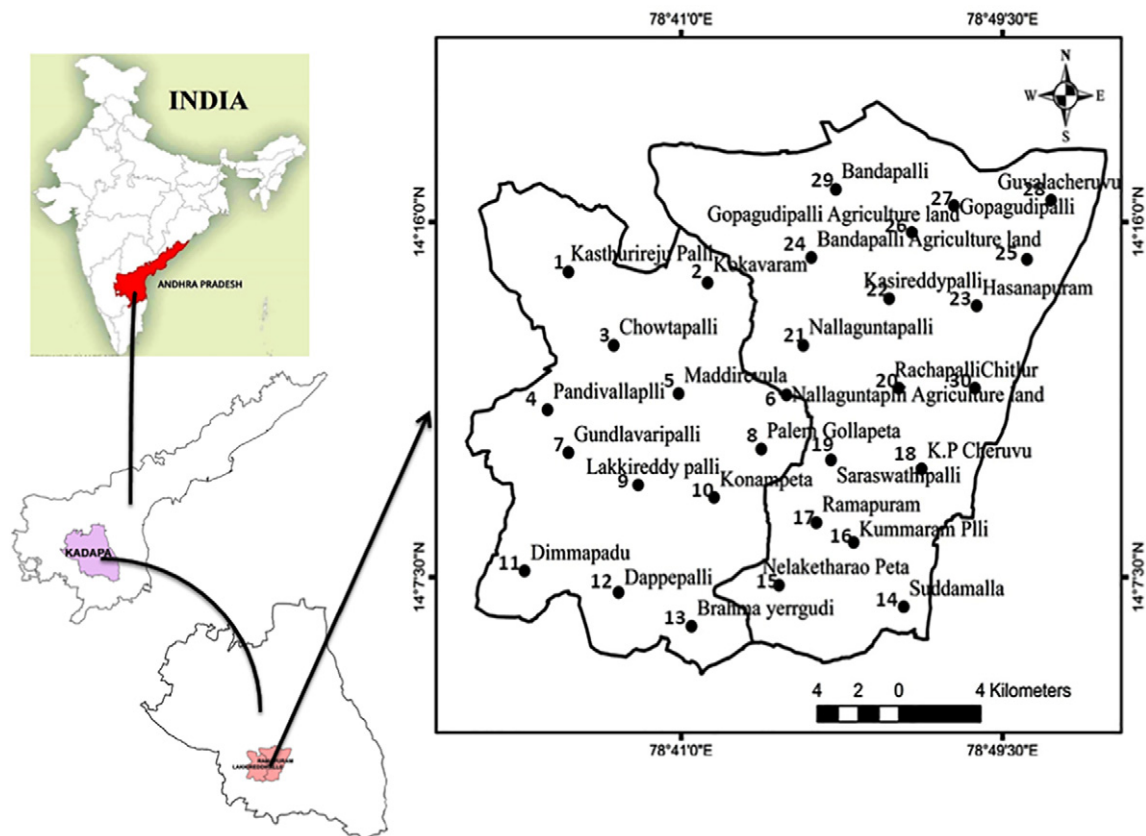


Fig. 1. Location map of the study area.

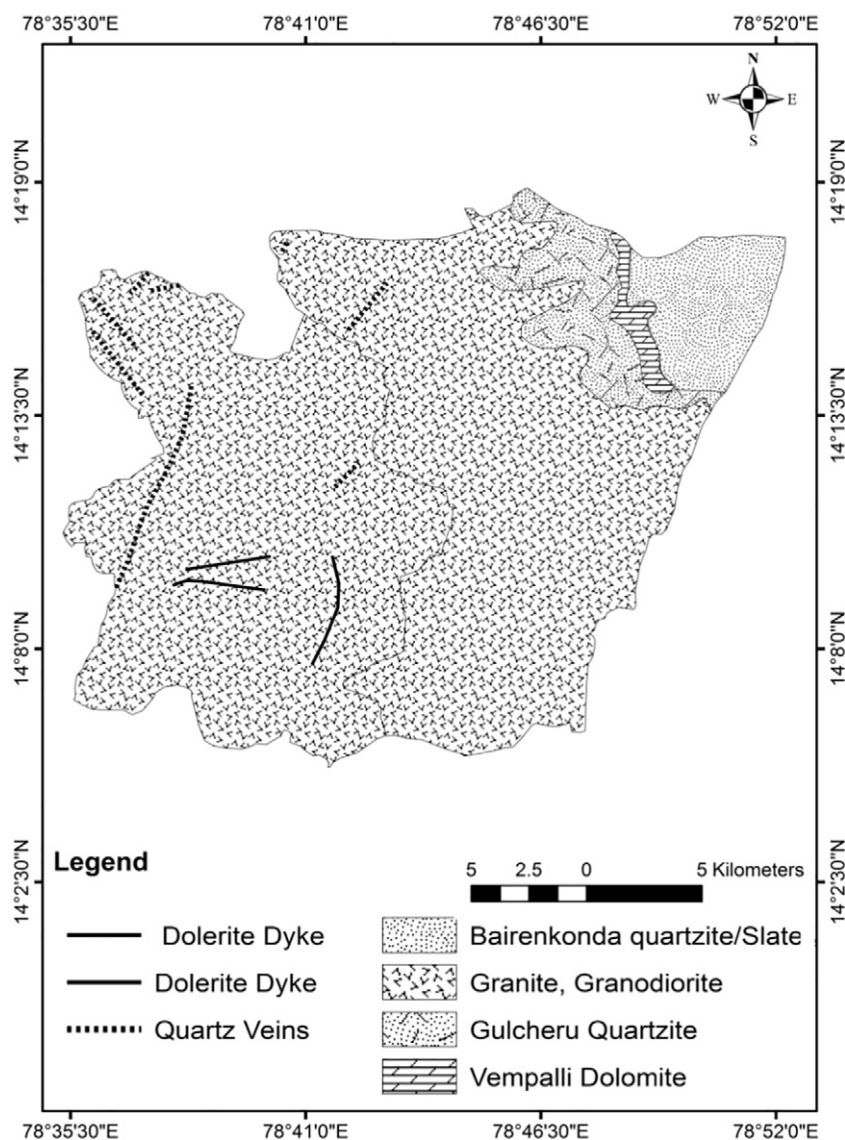


Fig. 2. Geology map of study area.

4. Results and discussion

pH is important in classifying many types of geochemical equilibrium. pH of groundwater ranges from 7.8 to 8.7 indicating alkaline nature. pH values for all the samples are within the desirable limits (WHO, 2011). Water with a pH higher than 8.5 or lower than 6.5 can produce aesthetic effects such as staining and etching or scaling of equipment. Electrical conductivity (EC) of the groundwater varies between 1200 $\mu\text{S}/\text{cm}$ –3800 $\mu\text{Siemens}/\text{cm}$ at 25 °C which accounts for a higher amount of salts in groundwater. Higher EC may be due to high mineral content and increased salinity at the studied location, temperature, types of ions, the concentration of EC increases with the concentration of TDS (Hem, 1985). EC can be categorized as type I, if the enrichment of salts are low ($\text{EC} < 1500 \mu\text{S}/\text{cm}$); type II, if the enrichment of salts are medium ($\text{EC} 1500$ and $3000 \mu\text{S}/\text{cm}$); and type III if the enrichment of salts are high ($\text{EC} > 3000 \mu\text{S}/\text{cm}$). Based on the above classification of EC, 23.33% of samples belongs to type I (low enrichment of salts) and 53.33% of samples fall in type II (medium enrichment of salts) and another 23.33% of samples belong to type III (high enrichment of salts). Higher variations in electrical conductivity might also be due to geochemical processes and anthropogenic activities in this region (Subba Rao et al., 2017; Narsimha and Sudarshan, 2013). Total Dissolved

solids vary from 650 mg/L to 2520 mg/L (Tables 1, 2) and 53.33% of samples are exceeding the permissible limit and are not potable. Total dissolved solids in water comprise all inorganic salts that include carbonates, bicarbonates, chlorides, fluoride, sulphates, nitrates, calcium, magnesium sodium and potassium (Sawyer, 1994). According to US Geological Survey (2000), 23.33% of groundwater samples can be grouped into freshwater category, whereas 76.66% of groundwater samples belong to slightly saline category (Table 3). High concentration of TDS can be due to anthropogenic sources such as percolation of domestic sewage into groundwater, leaching of salts from soil and also from septic tanks and agricultural activities (Krishna Kumar et al., 2017). Total hardness varies from 30 mg/L to 560 mg/L. Groundwater with TH < 75, 75–150, 150–300, >300 mg/L designated as soft, moderately hard, hard and very hard respectively (Sawyer and McCarthy, 1967) (Table 3). Based on this 33.33% of the samples belong to moderate hard and hard classification. Calcium and magnesium have contributed to groundwater hardness. Carbonate (CO_3^{2-}), bicarbonate (HCO_3^-) in groundwater varies from 14 to 40 mg/L and 50–440 mg/L with a median value of 23 and 184 mg/L, respectively (Tables 1, 2). Most of the groundwater samples are within the prescribed limit of 300 mg/L (WHO, 1990; BIS, 2012, Table 2). Chlorides in groundwater ranges from 52 to 960 mg/L with a median value of 292 mg/L (Table 2). 50% of water is

Table 1

Analytical data of the groundwater samples of the Lakkireddypalli, Ramapuram region.

S.No	Latitude	Longitude	pH	EC	TDS	TH	Na ⁺	K ⁺	Ca ²⁺	Mg ²⁺	CO ₃ ²⁻	HCO ₃ ⁻	TA	Cl ⁻	SO ₄ ²⁻	NO ₃ ⁻	F ⁻
1	N14°28'24"	E78°42'45"	8.3	3600	2400.2	150.2	130.2	99.7	40.1	49.8	40.1	73	112.8	130.0	189.6	50.4	1.1
2	N14°17'01"	E78°44'28"	8.2	2600	1800.4	190.1	179.6	139.6	50.4	34.3	28.9	98.1	126.2	99.8	180.1	65.1	3.4
3	N14°16'21"	E78°43'24"	8.7	1300	820.2	99.8	139.7	80.4	22.4	40.2	38.8	299.8	338.4	230.2	219.7	59.8	2.5
4	N14°16'39"	E78°41'21"	8.4	1560	949.7	139.8	280.4	170.1	29.8	22.2	18.7	179.6	198.2	80.4	160.5	42.4	2.4
5	N14°14'37"	E78°42'54"	8.6	2100	1319.6	30.2	229.8	199.8	24.2	20.4	18.6	219.7	237.8	93.2	170.1	47.6	3.0
6	N14°14'49"	E78°44'49"	8	1200	649.8	380.3	260.4	190.4	70.1	24.4	20.1	270.4	290.2	760.1	142.4	80.4	2.4
7	N14°14'44"	E78°44'16"	8	2600	1620.3	479.8	189.7	170.2	210.2	23.2	21.8	130.5	152.4	490.1	110.2	42.6	1.4
8	N14°13'18"	E78°44'51"	8.2	3040	1819.8	319.7	199.9	190.1	60.4	16.4	23.7	220.2	244.4	669.8	150.1	30.4	2.5
9	N14°13'7"	E78°43'1"	8.6	2500	1509.7	98.8	209.7	179.6	70.2	20.2	26.4	180.1	206.5	159.7	189.7	40.1	0.8
10	N14°21'44"	E78°46'9"	8.6	1400	820.3	119.9	320.4	210.4	50.4	24.4	14.4	339.7	454.1	220.1	159.7	55.4	0.6
11	N14°10'53"	E78°45'30"	8	2600	1640.4	520.3	299.6	229.7	109.8	79.8	18.2	49.9	68.4	960.4	170.2	62.1	1.4
12	N14°09'53"	E78°45'8"	8.1	2800	1629.9	499.8	520.4	340.4	64.8	69.4	19.7	74.4	94.2	710.2	159.8	80.4	1.2
13	N14°08'57"	E78°45'24"	8.2	1300	829.8	439.7	420.2	290.4	44.6	90.4	22.4	200.3	221.7	399.8	160.1	70.4	0.7
14	N14°07'36"	E78°45'49"	8.4	2800	1699.7	150.3	499.6	399.9	55.7	70.2	13.8	180.7	194.2	220.3	130.3	76.2	0.9
15	N14°05'50"	E78°45'47"	8.4	2600	1529.6	189.7	380.4	210.4	96.4	60.2	16.4	339.7	456.4	339.7	180.5	80.4	0.8
16	N14°10'04"	E78°42'27"	8	1620	1100.3	340.3	330.2	240.4	150.2	50.1	18.1	52.4	70.4	520.2	160.4	29.4	0.2
17	N14°03'38"	E78°84'39"	7.8	3800	2400.2	559.6	429.7	299.7	50.2	70.7	25.2	78.5	103.2	52.4	129.7	63.4	0.8
18	N14°10'36"	E78°42'45"	8.2	1420	829.4	169.8	419.8	320.4	40.4	49.7	22.2	200.2	222.4	64.4	162.5	70.4	2.2
19	N14°10'18"	E78°40'38"	8.4	2600	1729.6	179.7	310.4	290.4	29.8	60.4	20.4	179.8	198.4	93.2	99.9	73.8	2.4
20	N14°12'34"	E78°39'04"	8.4	3800	2520.1	240.3	299.8	190.4	40.5	30.1	18.3	219.8	238.2	170.5	90.4	80.4	1.4
21	N14°10'51"	E78°37'14"	8.5	1520	1100.4	70.2	210.4	110.4	40.3	40.5	24.4	109.7	134.4	230.4	195.1	48.5	1.2
22	N14°13'37"	E78°37'47"	8.4	3620	2300.2	320.3	240.5	180.4	69.8	60.5	29.6	150.2	180.1	120.1	120.5	49.7	0.7
23	N14°13'30"	E78°39'02"	8.6	1700	1229.7	149.8	409.6	199.6	71.9	82.4	28.4	170.4	198.3	320.4	110.4	120.4	2.0
24	N14°13'47"	E78°40'20"	8.2	2600	1720.4	279.7	179.6	160.4	60.1	84.1	26.5	140.2	166.2	309.7	170.2	99.6	2.2
25	N14°13'43"	E78°84'30"	8.2	3600	2229.6	169.8	230.4	199.8	56.5	95.8	27.8	142.1	170.4	93.2	220.2	90.4	1.4
26	N14°13'42"	E78°41'57"	8.6	1800	1200.2	240.2	240.4	130.4	60.4	23.6	30.4	220.4	250.1	78.4	130.5	85.2	1.2
27	N14°13'42"	E78°42'39"	8.6	1330	829.8	149.9	199.8	130.5	49.8	27.8	22.4	270.5	291.8	122.1	89.4	80.4	2.1
28	N14°13'37"	E78°42'45"	8.2	3600	2299.1	169.7	190.4	120.2	79.8	40.1	26.4	239.8	265.8	130.4	199.9	60.4	0.8
29	N14°14'14"	E78°42'38"	8.3	1400	1020.3	299.8	229.7	199.7	76.4	30.2	22.2	98.2	120.1	350.1	170.4	70.2	2.2
30	N14°14'17"	E78°42'44"	8.6	1730	1120.5	390.3	340	199.8	82.4	62.2	28.4	220.1	248.2	520.4	180.2	50.5	0.6

(*All parameters are expressed in mg/L; EC in $\mu\text{S}/\text{cm}$ and pH as no units).

above the standard limit of 200 mg/L (WHO, 1990; BIS, 2012). Higher chlorides is generally considered as an indicator for pollution and acts as a tracer for groundwater contamination (Loizidou and Kapetanios, 1993). Chloride in groundwater may also be derived from various sources like weathering, leaching of rocks and soil, domestic and municipal effluents (Sarath Prashanth et al., 2012). According to Shand (1952) residual water in pores of granites may contain chloride. Leaching of upper soil layers by domestic, municipal, industrial wastes, dry climate are also the contributing factors for higher chloride concentration (Subba Rao et al., 2017). Higher chlorides in drinking water tastes salty and laxative effect in people not accustomed to it. Sulphates in the study area ranges from 90 to 220 mg/L with an average value of

150 mg/L indicating most of the groundwater lie within the desirable limit of 200 mg/L. Sodium in groundwater varies from 130 to 520 mg/L (Tables 1, 2). 21% of groundwater is above the prescribed limits of 200 mg/L (WHO, 1990; BIS, 2012). The contributing factors for increased Na⁺ are due to leaching of deposit and decomposition of various minerals. Water in contact with igneous rocks dissolves sodium from its natural source. Sodium may also enter natural waters through industrial, municipal waste discharges and runoff from diffuse sources (Reddy Muralidhara et al., 2019). The relative high concentration of sodium may adversely deform soil structure and permeability, resulting in alkaline soils. Excessive amounts of sodium in drinking water normally affects the potability of water, and water containing up to 1000 mg/L

Table 2

Comparison of the quality parameters of groundwater samples of the study area with WHO standards.

Parameter	Expressed	Minimum	Maximum	WHO (2011)		BIS (2012)		Sample numbers exceeding the permissible limit (WHO, 2011)
				Desirable Limit	Permissible Limit	Desirable Limit	Permissible Limit	
pH	Units	7.8	8.7	6.5–8.5	9.2	6.5	8.5	Nil
EC	$\mu\text{S}/\text{cm}$	1200	3800	750	1500	–	–	1,2,4,5,7,8,9,11,12,14,15,16,17,19,20,21,22,23,24,25,26,28
TDS	mg/L	649.8	2520.1	500	1500	500	2000	1,2,18,9,11,12,14,15,17,19,20,21,22,23,24,25,26,28
TH	mg/L	30.2	559.6	100	500	–	–	11,7
Na ⁺	mg/L	130.2	520.4	50	200	–	–	4,5,6,9,10,11,12,13,14,15,16,17,18,19,30
K ⁺	mg/L	80.4	399.9	–	–	–	–	–
Ca ²⁺	mg/L	22.4	210.2	75	200	75	200	7
Mg ²⁺	mg/L	16.4	95.8	–	–	300	100	Nil
CO ₃ ²⁻	mg/L	13.8	40.1	–	–	–	–	Nil
HCO ₃ ⁻	mg/L	49.9	339.7	300	–	–	–	Nil
TA	mg/L	68.4	456.4	–	–	–	–	Nil
Cl ⁻	mg/L	52.4	960.4	250	600	250	1000	6,9,11,12,30
SO ₄ ²⁻	mg/L	89.4	220.2	2.5	375	200	400	Nil
NO ₃ ⁻	mg/L	29.4	120.4	–	45	45	No relaxation	1,2,3,5,6,7,8,9,10,11,12,14,15,18,19,20,21,22,23,24,25,26,27,28,29,30
F ⁻	mg/L	0.2	3.4	0.5	1.5	1	1.5	2,3,4,5,6,8,18,19,23,24,27,29

TA – Total Alkalinity, EC – Electrical conductivity, TDS – Total dissolved solids, TH – Total Hardness.

Table 3
Groundwater classification based on irrigational water quality parameters.

Irrigation suitability					
Parameter	Classification	Range	No. of Samples	% of Samples	Reference
TDS	Fresh water	<1000	7	23.33	US Geological Survey (2000)
	Slightly saline	1000–3000	23	76.66	
	Moderately saline	3000–10,000	Nil	Nil	
	High saline	10,000–35,000	Nil	Nil	
TH	Safe	<75	1	0.03	Sawyer et al. (1967)
	Moderate hard	75–150	10	33.33	
	Hard	150–300	10	33.33	
	Very hard	>300	9	0.3	
%Na	Excellent	>20	Nil	Nil	Eaton (1950)
	Good	20–40	Nil	Nil	
	Permissible	40–60	4	13.33	
	Doubtful	60–80	21	70	
KR	Unsuitable	>80	5	16.66	Kelly (1940)
	Suitable	<1	4	13.33	
	Unsuitable	>1	26	86.66	
	Suitable	>75%	14	47	
PI	Good	25–75%	16	53	Doneen (1964)
	Unsuitable	<25%	Nil	Nil	
	Suitable	<50	10	33.33	
	Unsuitable	>50	20	66.66	
PS	Suitable	<3	6	20	Doneen (1962)
	Unsuitable	>3	24	80	
	Excellent	<10	30	60	
	Good	10–18	Nil	40	
SAR	Doubtful	18–26	Nil	Nil	Richards (1954)
	Unsuitable	>26	Nil	Nil	
	Good	<1.25	30	100%	
	Doubtful	1.25–2.5	Nil	Nil	
RSC/RA	Unsuitable	>2.5	Nil	Nil	Eaton (1950)
	Suitable	<2.5	Nil	Nil	

may generally be physiologically tolerable (Ramesh and Anbu, 1996). However increased values of sodium in drinking water cause hypertension, arteriosclerosis, oedema and hyperosmolarity, kidney, nervous disorders in human body (Narsimha and Sudharshan, 2017). High sodium is not suggestive for agricultural practices as it depreciates the soil (Sarath Prashanth et al., 2012). The source of sodium (Na^+) is derived geologically from leaching of deposit and decomposition of various minerals. Potassium concentration of the groundwater is ranging from 80 to 400 mg/L with a median of 203 mg/L. The prescribed limit of K^+ for drinking water is noted as 12 mg/L as per WHO (1990) and all the samples are above this permissible limit. Calcium and Magnesium often are vital component of bone and nervous system etc. and they also influence the metabolic activities of the body. Calcium in the groundwater varies from 22 to 210 mg/L with a mean of 65 mg/L (Table 2). The permissible limit of calcium in drinking water is 75 mg/L (WHO, 1984). Only 3% of groundwater is above the standard limit of calcium. The study area was found to be calcium deficit where only 3% had shown calcium enrichment beyond the maximum desirable limit of 75 mg/L, as prescribed by BIS, 2012 for drinking purpose. Insufficient intakes of calcium may cause osteoporosis, defective teeth, nephrolithiasis (kidney stone), rickets, hypertension and stroke etc. (Faruqi, 2002). Magnesium values varied between 16 and 96 mg/L (Tables 1, 2). The required standard limit of magnesium in groundwater for drinking purpose is 30 mg/L (WHO, 1990; BIS, 2012). Most of the samples are within the standard limit of magnesium. Nitrates in groundwater varies from 30 mg/L–120 mg/L. 17% of the groundwater samples has permissible limit (45 mg/L) of nitrate concentration according to WHO standards. Higher nitrates concentration due to agricultural activities, septic tank leakage, domestic sewage (Datta and Tyagi, 1996; Sunitha et al., 2012a). High concentration of nitrates not only causes

methemoglobinemia in infants and also causes cancer, birth malformations gastric cancer, goiter, hypertension (Dissanayake et al., 1987; Bao et al., 2017; Fan, 2011). Children and young livestock are more prone to nitrate contamination in drinking water. Though methemoglobinemia (blue-baby syndrome) is the deadliest effect of nitrate exposure, there are other serious chronic health impacts like thyroid enlargement, 15 types of cancers, birth defects, hypertension etc. (Forman et al., 1985). Most of the people in this area rely on agricultural practices for their livelihood, application of fertilizers for crop yields could be the main contributing factor for high nitrate concentration. In addition to the agricultural practices, poultry waste disposal and household/cultivate animal dung contribute higher nitrate concentration are leached in groundwater (Raju et al., 2009 Lokesh, 2013; Oenema et al., 2005; Adimalla et al., 2018a; Sunitha et al., 2012c). Fluoride is good for human health in preventing dental caries and helps in enamel development at 0.8–1.0 mg/L for children below 8 years of age (Sunitha and Muralidhara Reddy, 2018). If the groundwater contains in excess of 1.5 mg/L, it causes dental fluorosis & skeletal fluorosis if it is found in excess of 3.0 mg/L if that water is consumed consistently for 8–10 years. High manifestations of dental fluorosis are mostly found in children up to the age of 12 years. Dental fluorosis is observed in parts of Vemula, Y.S.R District, Kadapa; in parts of Kadapa, Anantapur District (Sunitha et al., 2013a, 2013b); in Badvel, Y.S.R District (Sunitha et al., 2014b, 2014a) and also in and around abandoned mines, Vemula, Y.S.R District (Reddy Sudarshan et al., 2018). In this study region highest concentration of fluoride 2.93 mg/L is observed at Kasireddypalli and lowest concentration of fluoride 0.25 mg/L is observed at Ravindranagaram. Fluoride concentration (1.5–3.0 mg/L) is observed in Kasireddipalli, Gopagudipalli, Mulapalli, Kalpanayenicheruvu, Medharapalli villages. The study reveals that children are highly prone to the health risks caused by dental fluorosis through the intake of elevated fluoride water. Principal source of fluoride in the groundwater is fluoride bearing rocks. Weathering of host rocks wash out fluoride and contaminate the ground water. Fluorides occurs in three forms, namely fluorspar or calcium fluoride (CaF_2), apatite or rock phosphate ($\text{Ca}_3\text{F}(\text{PO}_4)_3$) and cryolite (Na_3AlF_6) (Sunitha et al., 2012a). Possible source of fluoride in groundwater are weathering and leaching of fluoride bearing minerals from the basement granitic rocks of this region under alkaline environment. The Geological formation is found to be a main source for the fluoride concentration in most of the sampling points. In addition to this, seasonal, spatial and depth variations are also the possible causes for variations of fluoride concentration in groundwater (Sunitha et al., 2012b; Sunitha and Muralidhara Reddy, 2018).

4.1. Gibbs diagram

Gibb's diagram is plotted in order to assess the mechanism controlling groundwater chemistry of Lakkireddi palli region. $\text{Na}^+ + \text{K}^+ / \text{Na}^+ + \text{K}^+ + \text{Ca}^{2+}$ and $\text{Cl}^- / \text{Cl}^- + \text{HCO}_3^-$ as a function of TDS are plotted to denote precipitation dominance (PD), rock dominance (RD) and evaporation dominance (ED). All groundwater samples were plotted as shown in Fig. 3a & b. From the figures it is clear that all the groundwater samples were in the rock dominance (RD) zone revealing rock weathering and dissolution of fluoride bearing minerals from the source rock are the chief contributing factors for release of fluoride to groundwater (Sunitha et al., 2012a; Narasimha and Sudharshan, 2017; Adimalla and Venkatayogi, 2018; Faten et al., 2016; Thomas et al., 2016).

4.2. Normalized bivariate plots

A number of lithological processes including weathering and dissolution control the groundwater chemistry. It is evident by observing the plots of Ca/Na versus HCO_3^-/Na and Ca/Na versus Mg/Na (Fig. 4a, b) groundwater chemistry depends on silicate weathering and partially by carbonate dissolution. Therefore sodium potassium, calcium and magnesium are greatly contributed through silicate weathering. Similar

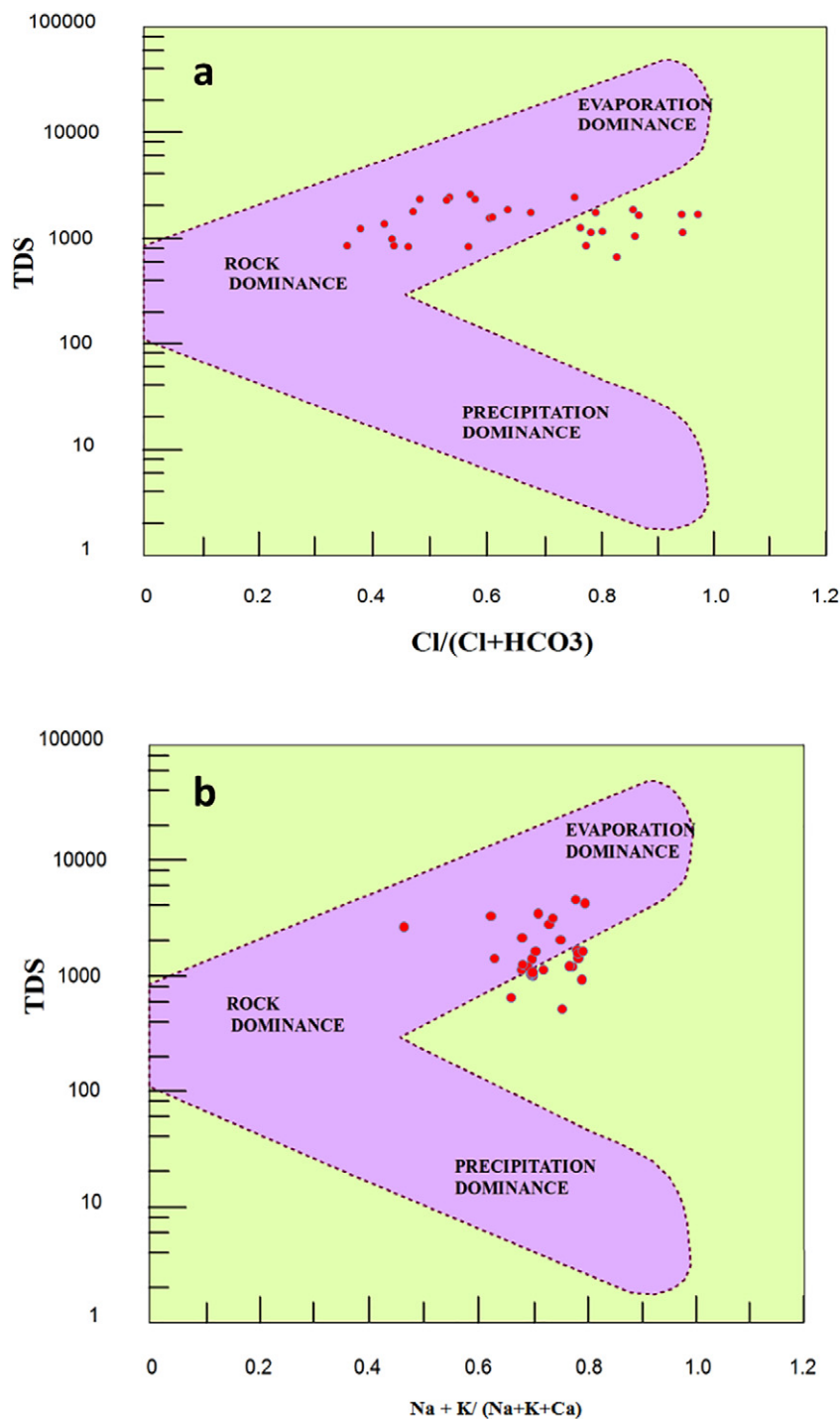


Fig. 3. Mechanism controlling groundwater chemistry for LR Palli, Ramapuram region a. TDS vs. $(\text{Cl}^-) / (\text{Cl}^- + \text{HCO}_3^-)$, b TDS vs. $(\text{Na}^+ + \text{K}^+) / (\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+})$.

findings were noticed in silica rich lithogenic areas (Brindha et al., 2016; Rajesh et al., 2012; Kumar, 2014).

4.3. Hydro-geochemical facies

Hydro geochemical regime of groundwater is represented by plotting major cations and anions in the trilinear diagram (Piper, 1953). In the present study region groundwater types can be grouped in to six types like CaHCO_3 , Na Cl , Mixed CaNaHCO_3 , Mixed Ca Mg Cl , CaCl , and Na HCO_3 . Based on the Piper's plot (Fig. 5) majority of the samples belongs to Mixed Ca Mg Cl type.

4.4. Irrigation suitability

To assess the suitability of groundwater with respect to agricultural/irrigation practices the following methods like sodium percentage (% Na; Wilcox, 1955), Sodium adsorption ratio (SAR; Richards, 1954); Residual sodium carbonate (RSC; Eaton, 1950; Raghunath, 1987), Magnesium hazard Ratio (MHR; Raghunath, 1987), Kelly ratio (KR; Kelly, 1940) and permeability Index PI; Doneen (1964) are formulated.

4.4.1. Percent sodium (%Na)

Percent sodium plays a crucial role in evaluating the suitability for irrigation. Sodium makes bonding with soil to reduce its permeability

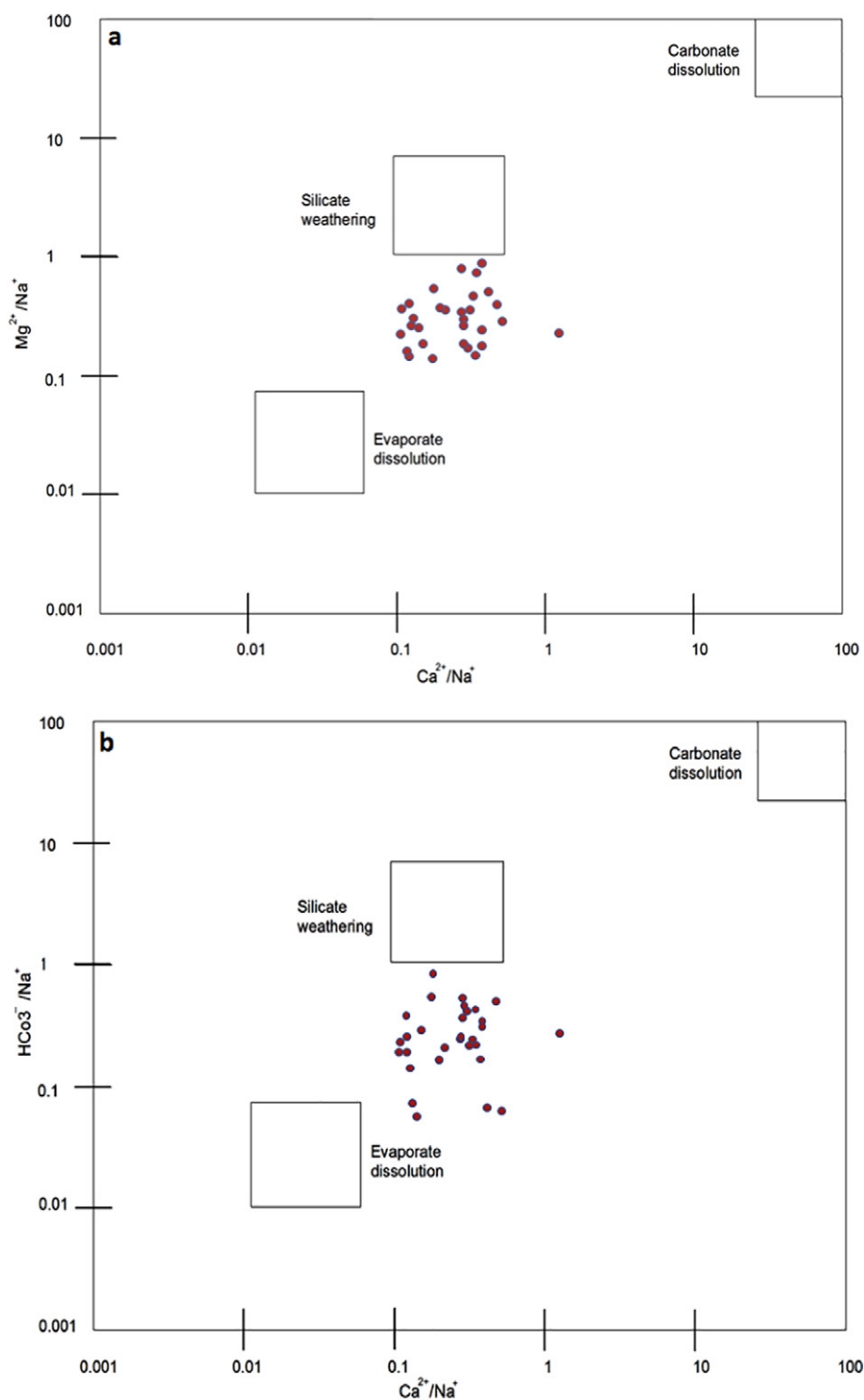


Fig. 4. Bivariate plots a. $\text{Ca}^{2+}/\text{Na}^+$ vs. $\text{Mg}^{2+}/\text{Na}^+$ b. $\text{Ca}^{2+}/\text{Na}^+$ vs. $\text{HCO}_3^-/\text{Na}^+$.

(Wilcox, 1955; Ayers and Westcot, 1985). Sodium reacts with carbonate forms alkaline soils, whereas sodium reacting with chloride forms saline soils. Plant growth is hindered with this type of soil (Todd, 1980). Higher levels of sodium in irrigation water tends the ions to mobilize towards particles of clay by omission of Ca^{2+} and Mg^{2+} ions through a base exchange reaction. This base exchange process in the soil reduces water movement capacity thereby restricting water during wet conditions, as a result such soils become hard dry. Sodium is generally

referred to as sodium percentage in irrigation water and can be denoted by the following formula (Wilcox, 1955).

$$\text{Percent Sodium} = \frac{\text{Na}^+ + \text{K}^+}{\text{Ca}^{2+} + \text{Mg}^{2+} + \text{Na}^+ + \text{K}^+} \times 100$$

Na, Ca, Mg and K are taken in meq/L. Based on the Wilcox classification around 40% of the groundwater belongs to doubtful unsuitable class

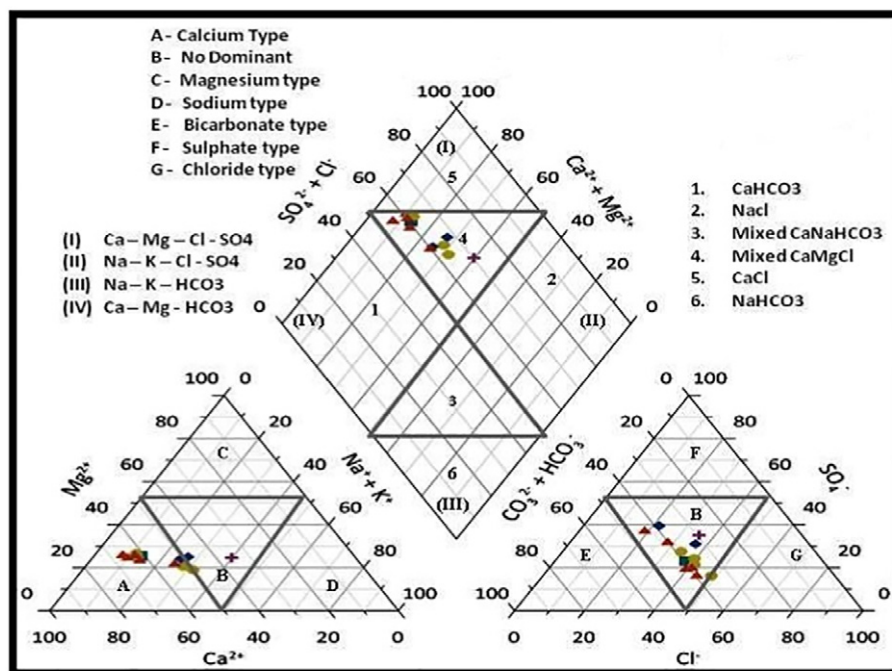


Fig. 5. Piper diagram.

(Fig. 6). It is observed that based on percent sodium, 70% of water is under doubtful class, 16.66% is under unsuitable class, 13.33% belongs to permissible class (Eaton, 1950). %Na ranges between 50 mg/L to 84 mg/L with the median value of 70 mg/L.

4.4.2. Sodium adsorption ratio (SAR)

Concentration of sodium is a significant parameter in the irrigation classification of water. Sodium reacts with soils in reducing its permeability. Sodium concentration is an important factor as its produces

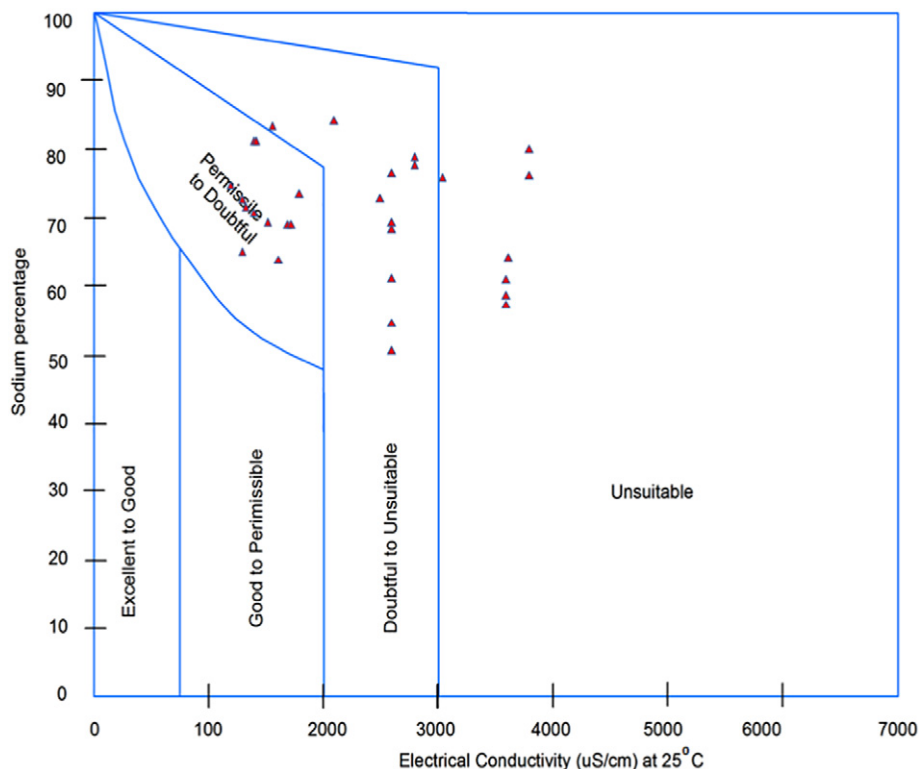


Fig. 6. Classification of groundwater quality with respect to electrical conductivity and percent sodium.

undesirable impacts of altering soil properties and decreases its permeability. Hence sodium concentration assessment solubility is crucial for the assessment of irrigation. Calcium and magnesium in soils makes soil more permeable and aids easy cultivation. Increased sodium in soils substitutes calcium and magnesium from them by making hard & compact causing to reduced infiltration and poor internal drainage (Karanth, 1987). The SAR (Sodium adsorption ratio) is determined by the equation given below:

$$SAR = \frac{[Na^+]}{\sqrt{\frac{[Ca^{2+} + Mg^{2+}]}{2}}}$$

where the concentrations are expressed in meq/L.

The greater the value of SAR, the greater is the hazard to the crops due to excess sodium. SAR values ranges from 0.77 to 2.38. 100% of the groundwater belongs to excellent category (Richards, 1954).

4.4.3. Classification by U.S Salinity Laboratory method (USSL)

Graphical method proposed by Raghunath (1987), for examining irrigation suitability of groundwater. It is essentially an adaptation of the

semi-logarithmic diagrammatic representation developed by the staff of U.S. Salinity Laboratory (1954), to suit Indian conditions. It consists of two parameters namely, (i) Electrical conductance (EC) of water expressed in $\mu S/cm$, which is proportional to TDS content and hence the degree of salinity and (ii) Sodium adsorption ratio (SAR).

In this EC is denoted as salinity hazard and SAR as an alkalinity hazard (Fig. 7) depicts 46.6% belongs to C3S1 class indicating high salinity and low alkalinity and 53.3% of samples fall into C4S1 indicates very high salinity and low alkalinity. The agricultural yields are generally low in lands irrigated with water belonging to doubtful to unsuitable category. This could be due to the presence of excess sodium salts which cause osmotic effects on soil-plant system.

4.4.4. Residual sodium carbonate (RSC)

RSC also influences the suitability of water for irrigation all over the world. RSC exists when carbonate and bicarbonate concentration exceeds calcium and magnesium content. Residual Sodium Carbonate can be determined by subtracting the alkaline earth ($Ca^{2+} + Mg^{2+}$) from the carbonate ($CO_3^{2-} + HCO_3^-$). When the sum of carbonates is in calcium and magnesium, there may be a possibility of complete precipitation of calcium and magnesium (Raghunath, 1987). Hence the relative proportion of sodium in water is increased in the form of sodium

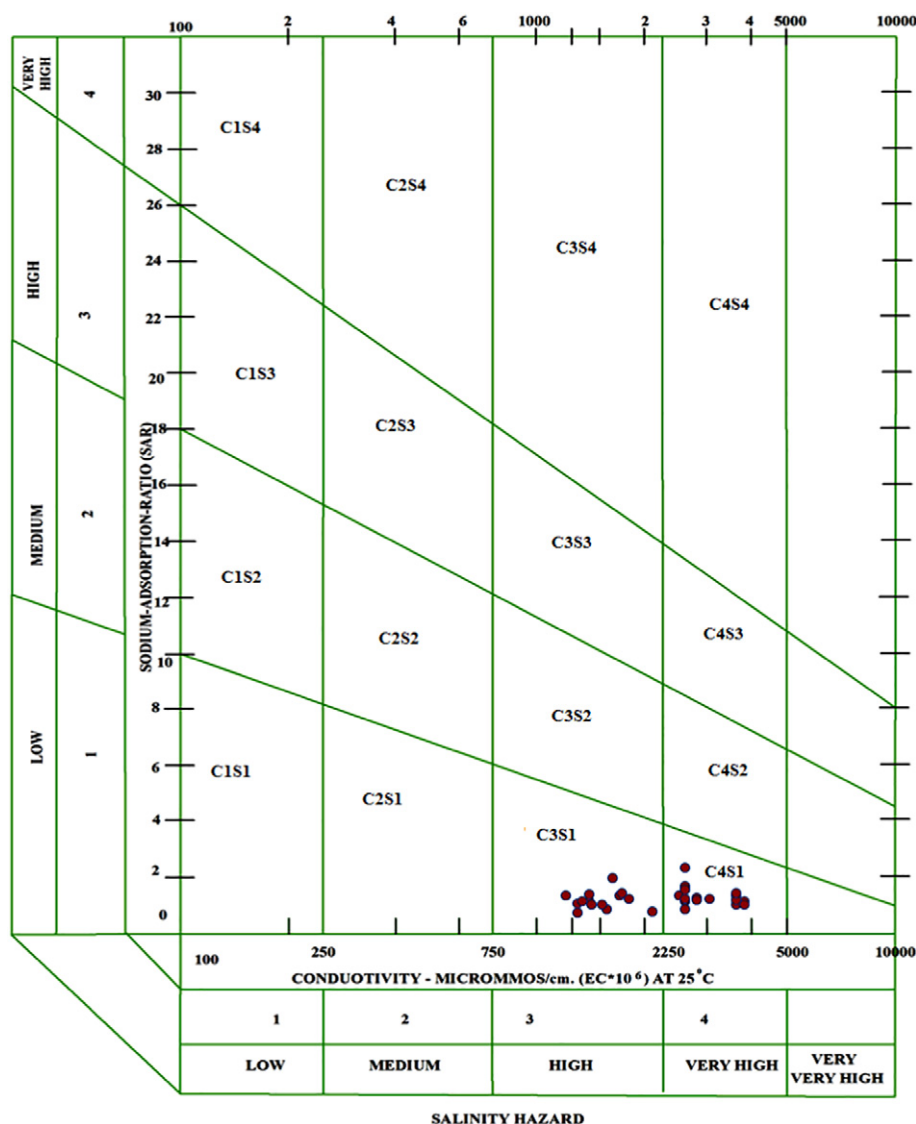


Fig. 7. USSL classification of groundwater.

carbonate and this is indicated by RSC, calculated as follows (Eaton, 1950; Raghunath, 1987).

$$\text{RSC} = (\text{CO}_3^{2-} + \text{HCO}_3^-) - (\text{Ca}^{2+} + \text{Mg}^{2+})$$

According to the above classification, water >2.5 meq/L of RSC is not suitable for irrigation. Based on RSC classification (Eaton, 1950). The RSC values in this region ranges between –10.65 to 3.21 (Table 3). 97% falls in the good category.

4.4.5. Magnesium Hazard (MH)

Calcium and Magnesium maintain a state of equilibrium in most waters a high level of Mg^{2+} is generally due to the presence of exchangeable Na^+ in irrigated soils. The excess Mg^{2+} alters the quality of soil making it alkaline ensuing decreased crop yield and poor agricultural returns. Mg hazard is determined by the equation given below:

$$\text{Magnesium Hazard (MH)} = \text{Mg}^{2+} \times 100 / (\text{Ca}^{2+} + \text{Mg}^{2+})$$

MR value <50 is suggested for irrigation while MR value >50 is not suggested for irrigation thereby decreasing the crop yield (Szabo and Darab, 1964). The Mg hazard in this region ranges between 15.29 and 76.72 (Table 3). 33.33% of groundwater samples are within Mg hazard while 66.66% of groundwater samples are unsuitable for irrigation.

4.4.6. Permeability index (PI)

Permeability Index is an important parameter for classifying irrigation water permeability of the soils and is inclined by prolonged application of irrigation water (higher salts) as it is influenced by Na^+ , Ca^{2+} , Mg^{2+} , and HCO_3^- ions in soils. Doneen (1964) developed permeability index PI and is determined by the equation given below:

$$\text{PI} = (\text{Na}^+ + \sqrt{\text{HCO}_3^-} \times 100) / (\text{Ca}^{2+} + \text{Mg}^{2+} + \text{Na}^+)$$

(where all the ions are expressed in meq/L).

PI values of this region range from 47.12 to 92.65. PI is divisible three classes. Class I (75 Suitable), Class II (25–75% good) and Class III (25% unsuitable). 47% of groundwater belongs to class I and 53% of groundwater belongs to class II category. Water under class I and class II is recommended for irrigation (Doneen, 1964).

4.4.7. Potential salinity (PS)

Potential salinity is applied for categorization of water for agricultural purpose. PS <3 meq/L is good for agricultural applications (Doneen, 1962). PS values of this region range from 2.8 to 28.8. 80% of the samples belong to the unsuitable class and only 20% of the groundwater samples are suitable for irrigation (Table 3). PS can be calculated by the following equation.

$$\text{PS} = \text{Cl}^- + 0.5 \times \text{SO}_4^{2+}$$

4.4.8. Soltan classification

Soltan (1999) classified groundwater into base –exchange indices (r_1) and meteoric genesis indices (r_2) as given below:

$$r_1 = (\text{Na}^+ - \text{Cl}^-) / \text{SO}_4^{2-}$$

$$r_2 = [(\text{K}^+ + \text{Na}^+) - \text{Cl}^-] / \text{SO}_4^{2-}$$

where Na^+ , K^+ , Cl^- , SO_4^{2-} concentrations are expressed in meq/L. If $r_1 < 1$ and $r_2 < 1$, groundwater is of $\text{Na}^+ - \text{SO}_4^{2-}$ and deep meteoric type, respectively whereas $r_1 > 1$ and $r_2 > 1$ shows the source are of $\text{Na}^+ - \text{HCO}_3^-$ and shallow meteoric type respectively (Table 3). As per

Soltan classification, 100% groundwater belongs to $\text{Na}^+ - \text{CO}_3^{2-}$, deep meteoric type.

4.4.9. Kelly's ratio

Kelly's ratio signifies higher content of sodium in water (Kelly, 1940). KR ratio also one of the vital parameters in classifying irrigation water. KR index is alkali hazard indicator. In case of high KR values gypsum is advised to decrease the effects of Na^+ ion (Paliwal, 1967).

KR is determined by using the formula.

$$\text{KR} = \text{Na}^+ / (\text{Ca}^{2+} + \text{Mg}^{2+})$$

KR values of the groundwater sample range from 0.66 to 3.68 to the average value of 1.88. Water within $\text{KR} < 1$ is suitable for irrigation while water within $\text{KR} > 1$ is unsuitable. From Table 3, 13.33% fall in a suitable class and 86.66% fall in unsuitable class for irrigation. It is clear that most of the water samples are not suitable for irrigation.

5. Conclusion

Groundwater is of alkaline, hard in nature and is exceeding the permissible limits of TDS, TH, EC, chloride, sulphates, fluorides and nitrates. Leaching of upper soil layers by domestic, municipal, industrial wastes, residual water in pores of granites might be the contributing factors for high chloride concentrations. Rock weathering and dissolution of fluoride bearing minerals from the source rock are the chief contributing factors for release of fluoride in groundwater. Groundwater chemistry depends on silicate weathering and partially by carbonate dissolution.

Dissolution of sulphate and chloride are the controlling factors of groundwater chemistry.

Dominant hydrogeochemical facies observed in this area is of Mixed Ca Mg Cl type. High concentration of nitrates is due to agricultural activities, leakage from septic tanks, and domestic sewage from this area. Groundwater suitability for irrigation reveals that most of the groundwater fall under unsuitable class for irrigation. Higher values of MH, KI, PS is observed in most of the locations. Application of this untreated water may be harmful for crops and further may decreases in its yield. Agricultural yields are low in lands irrigated with water belonging to doubtful and unsuitable category. However more saline tolerant crops could be suggested to combat soil salinization and application of lime or gypsum may be suggested to about the soil permeability. These types of studies will facilitate in identifying suitable sample location for irrigation systems and further prevent the damage of crops and reduced yield.

CRedit authorship contribution statement

V. Sunitha: Conceptualization, Formal analysis, Investigation, Supervision, Validation, Writing - review & editing. **Y. Sudharshan Reddy:** Conceptualization, Formal analysis, Validation, Writing - review & editing.

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